

CSE420: Compiler Design

Fall 2024

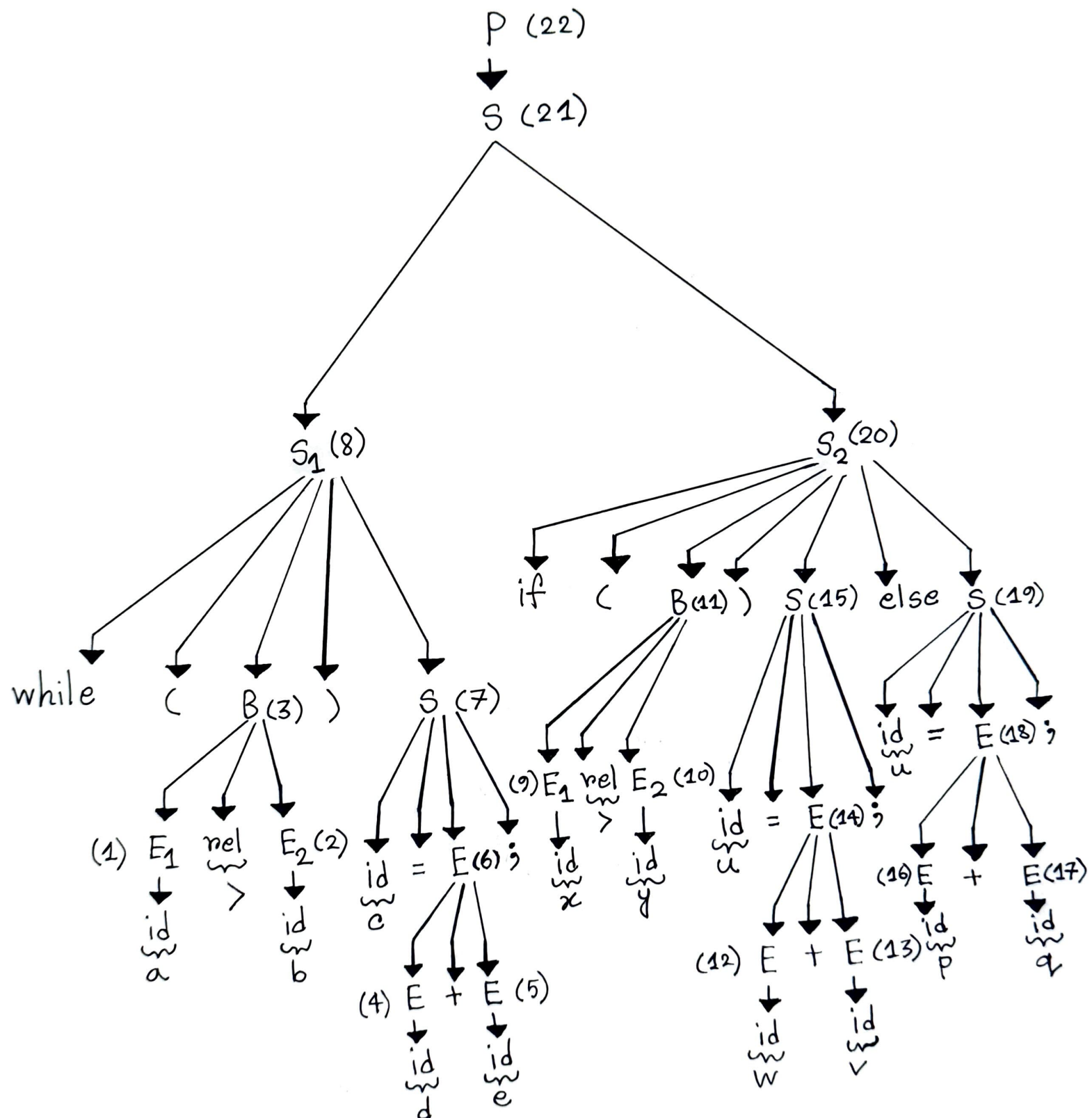
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Section: 06

Theory Faculty Initial: SZD

While (a > b) c = d + e; if (x > y) u = w + v; else u = p + q



- 1) E.add = a
E.code = "
- 2) E.add = b
E.code = "
- 3) B.true = L1
B.false = S.next = L4
B.code = if $a > b$ goto B.true
goto B.false [L4]

- 4) E.add = d
E.code = "
- 5) E.add = e
E.code = "
- 6) E.add = t_1
E.code = $t_1 = d + e$

- 7) S.next = begin
S.code = $t_1 = d + e$
 $c = t_1$

- 8) S.next = L4

begin = begin: if $a > b$ goto L1
goto S.next [L4]
L1:
S.code = $t_1 = d + e$
 $c = t_1$
goto begin

- 9) E.add = x
E.code = "

- 10) E.add = y
E.code = "

- 11) B.true = L2
B.false = L3
B.code = if $x > y$ goto B.true L2
goto B.false L3

- 12) E.add = w
E.code = "

- 13) E.add = v
E.code = "

- 14) E.add = t_2
E.code = $t_2 = w + v$

- 15) S.next = S.next = L5
S.code = $t_2 = w + v$
 $u = t_2$

- 16) E.add = p
E.code = "

- 17) E.add = q
E.code = "

- 18) E.add = t_3
E.code = $t_3 = p + q$

- 19) S.code = $t_3 = p + q$
 $u = t_3$
S.next = S.next = L5

21) $S.next = L5$

$S.code = begin = begin: if\ a > b\ goto\ L1$
 $goto\ S.next\ L4$
 $L1:$
 $S.code = t_1 = d + e$
 $c = t_1$
 $goto\ begin$
 $L4:$
 $if\ x > y\ goto\ L2$
 $goto\ L3$
 $L2:$
 $t_2 = w + v$
 $u = t_2$
 $goto\ S.next\ L5$
 $L3:$
 $t_3 = p + q$
 $u = t_3$

$P.code = begin = begin: if\ a > b\ goto\ L1$
 $goto\ L4$
 $L1:$
 $t_1 = d + e$
 $c = t_1$
 $goto\ begin$
 $L4:$
 $if\ x > y\ goto\ L2$
 $goto\ L3$
 $L2:$
 $t_2 = w + v$
 $u = t_2$
 $goto\ L5$
 $L3:$
 $t_3 = p + q$
 $u = t_3$
 $L5:$